HYDRAULIC LASH ADJUSTER (HLA) INSPECTION [SKYACTIV-D 1.5]

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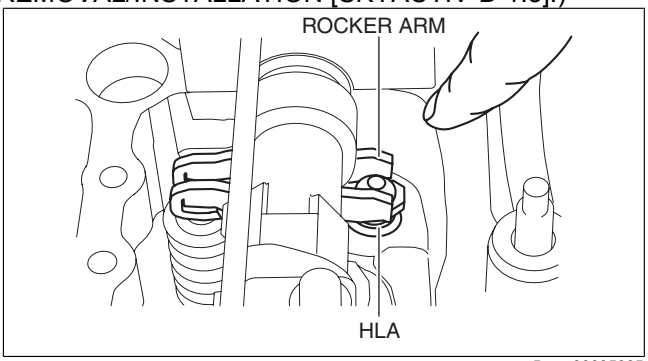
HLA Troubleshooting

Item	Occurrence	Possible cause	Action
1	After replacing oil, sound occurs when starting engine	Oil shortage in oil passage	• Maintain the engine speed between 2,000 to 3,000 rpm for approx. 2 s and 10 min*, and verify that the sound disappears. — If the sound disappears, the HLA is normal. — If the sound does not disappear, go to item 5.
2	Sound occurs at instant when engine is started on vehicle left for 1 day		
3	Starter cranked for 3 s or more and sound occurs directly after engine is started	HLA oil shortage	
4	Sound occurs directly after engine is started with newly replaced HLA		
5	Sound occurs continuously for 10 min or more	Insufficient oil pressure	• Inspect the oil pressure. (See OIL PRESSURE INSPECTION [SKYACTIV-D 1.5].) — If less than the specification, inspect each part of the engine.
		HLA malfunction	• Inspect the HLA. (See HLA Inspection.)
6	Sound occurs during idling after high-speed driving	Incorrect engine oil level	• Inspect the engine oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].)
		Engine oil deterioration	• Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].)

* : Period of time engine oil flows to the HLA, this value includes tolerance caused by the ambient temperature and the oil used.

HLA Inspection

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the fuel injectors. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Remove the cylinder head cover. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Press the rocker arm from above and verify that the HLA does not drop.
   - If the HLA drops, replace the HLA. (See HYDRAULIC LASH ADJUSTER (HLA) REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Install in the reverse order of removal.



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